Figure preview — shown at final printed size

Column width 82.5 mm, full width 170 mm. Grayscale only, so the figures stay legible in black and white as the journal requires. The text under each figure is the draft caption.

`fig1_concept` *full width*

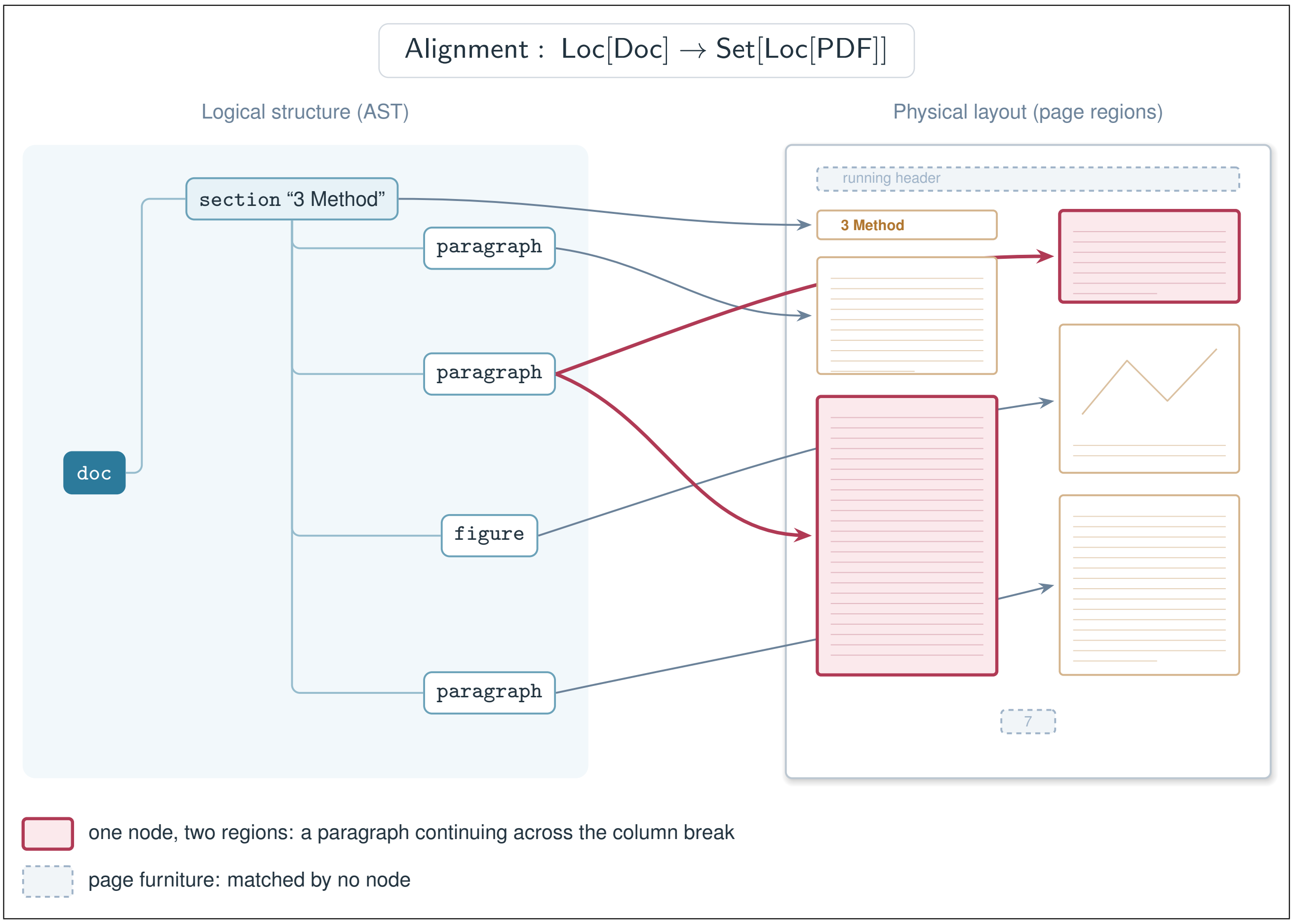

Figure 1: The two views of a document and the alignment between them. The logical structure carries type and hierarchy but no position; the page regions carry position but no structure. The mapping is not one-to-one: the second paragraph node covers two regions because the paragraph continues across the column break, while page furniture such as the running header and the page number is matched by no node at all.

`fig3_readingorder` *column width*

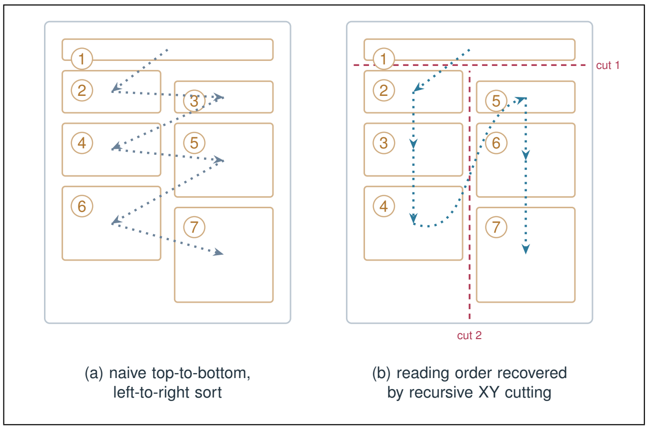

Figure 3: Detected regions do not arrive in reading order. (a) Sorting regions top-to-bottom and left-to-right interleaves the two columns, so the sequence of regions no longer corresponds to the sequence of nodes. (b) Recursive XY cutting splits the page at gaps that no region straddles, recovering the order a reader would follow.

`fig2_pipeline` *column width*

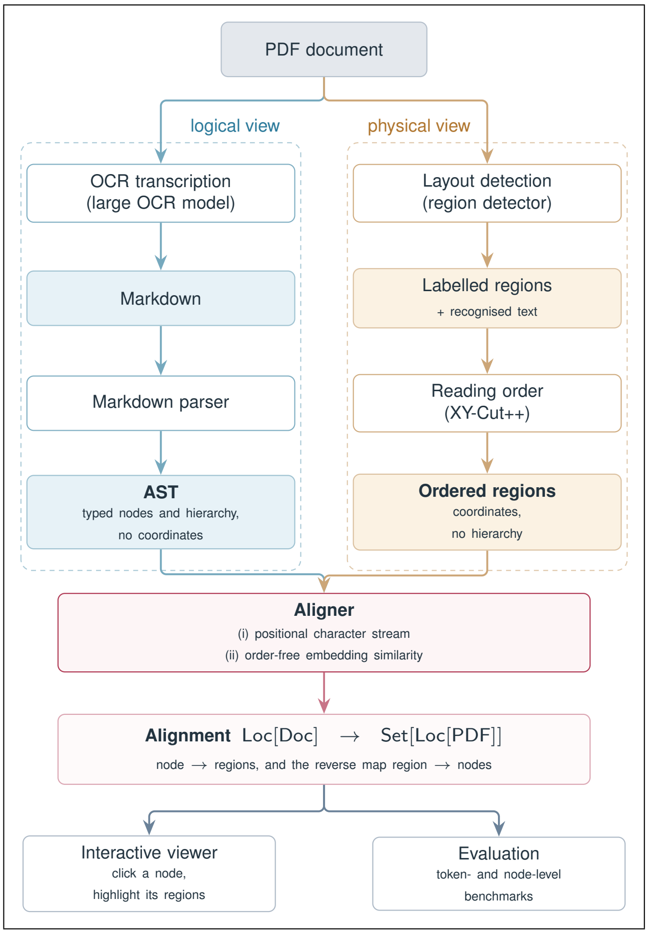

Figure 2: The system. Both views are built independently from the same PDF and share no coordinate system, no tokenisation and no unit of comparison. The aligner is the only component that sees both, and the alignment it produces drives the interactive viewer and the evaluation.

`fig4_aligners` *column width*

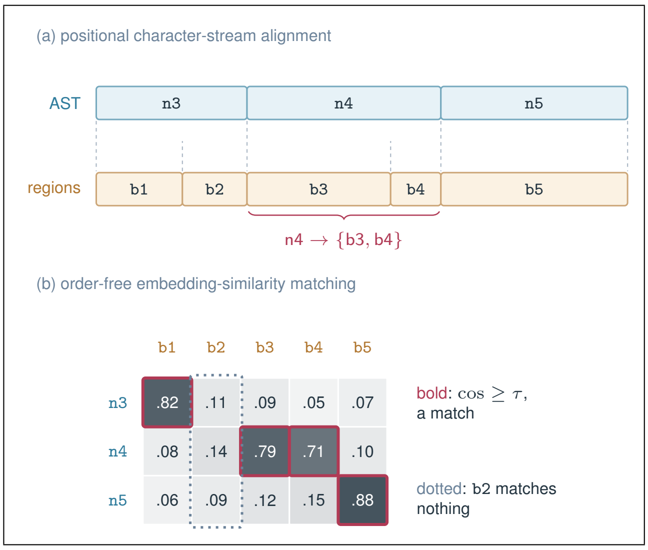

Figure 4: The two aligners rest on opposite assumptions. (a) The stream aligner flattens both sides into normalised character streams and diffs them, so a node is mapped by the position of its characters; matching is exact about repeated text but depends on the recovered reading order being right. (b) The similarity aligner scores every node against every region by embedding cosine and ignores order entirely; a region whose best score falls below the threshold τ, such as furniture, is left unmatched.

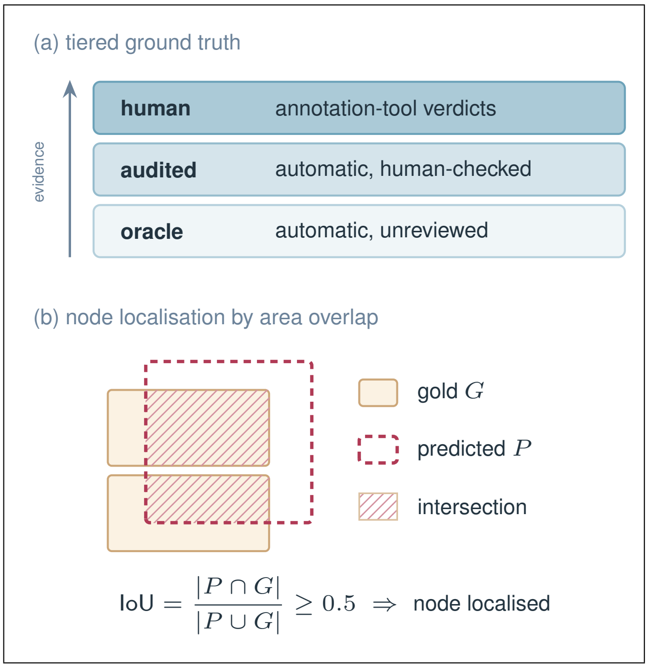

Figure 5: Evaluation. (a) Each node's ground truth is drawn from the strongest tier available, and every reported result states how much of it rests on each tier, so a number resting mostly on unreviewed automatic placement is labelled as such rather than presented as verified. (b) A node counts as localised when the pixel area its predicted regions cover overlaps the gold area at IoU ≥ 0.5. Areas rather than box identities are compared, so a node is not penalised when the detector splits its text into a different number of regions than the gold does.